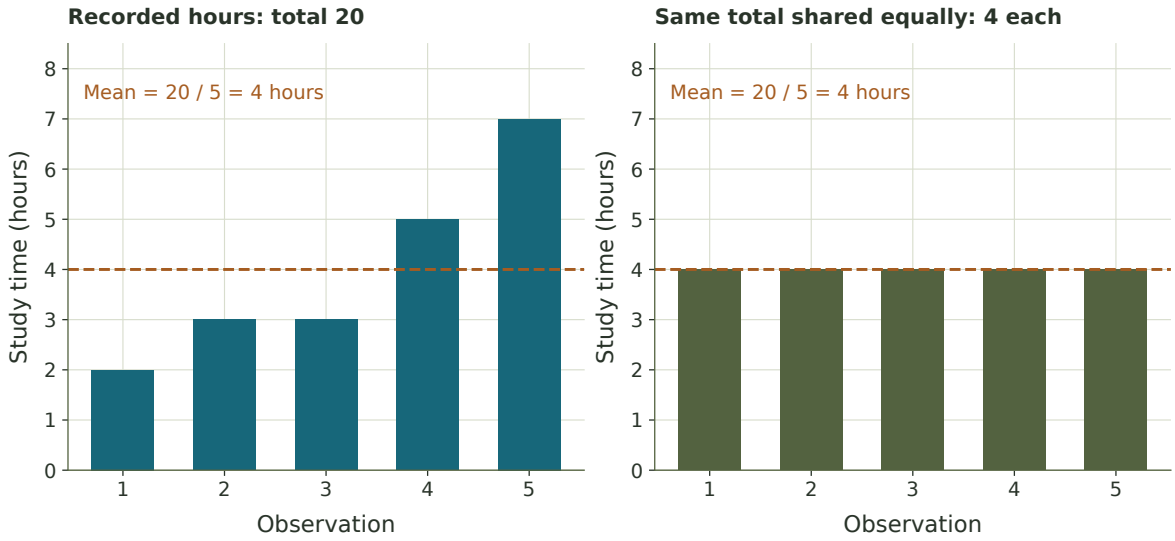
WEEK 3 / BASIC STATISTICS

Describing center and variation

Use numerical summaries and graphs to explain what a set of observations tells us.



A typical value is only part of the story

Two routes can average the same travel time while one varies much more from day to day. A useful description asks both where the observations sit and how much they differ. The setting, units, and graph help us interpret the calculations.

Week 2 organized observations with frequency tables, relative frequencies, and graphs. This chapter develops the mean, median, mode, range, and sample standard deviation. Weighted averages connect those methods to a numerical model of attention.

The mean distributes a total equally, the median identifies the middle position, and the standard deviation measures variation about the mean. The graph supplies context for those summaries.

Online reading · Center and spread explorer · Laboratory

3.1 What are the numbers about?

A list of measurements is a *data set*. Each number describes an observation: perhaps one day's waiting time or the amount in one bottle. Before calculating, ask what was measured, which units were used, and who or what the list describes.

A measure of *center* describes the typical location of the observations. A measure of *spread* describes how far apart they are. For a useful comparison, report both and consider the shape of the data.

The same average can describe different experiences

One route to work takes 24, 25, and 26 minutes. Another takes 15, 25, and 35 minutes. Both average 25 minutes. The second route varies more, with trips taking as long as 35 minutes. If arriving on time matters, you will want to know about that variation.

Units matter. Before averaging travel times, for example, convert all measurements to minutes or all to hours. Comparisons between different groups also require knowing how the observations were selected.

Pause and try 1

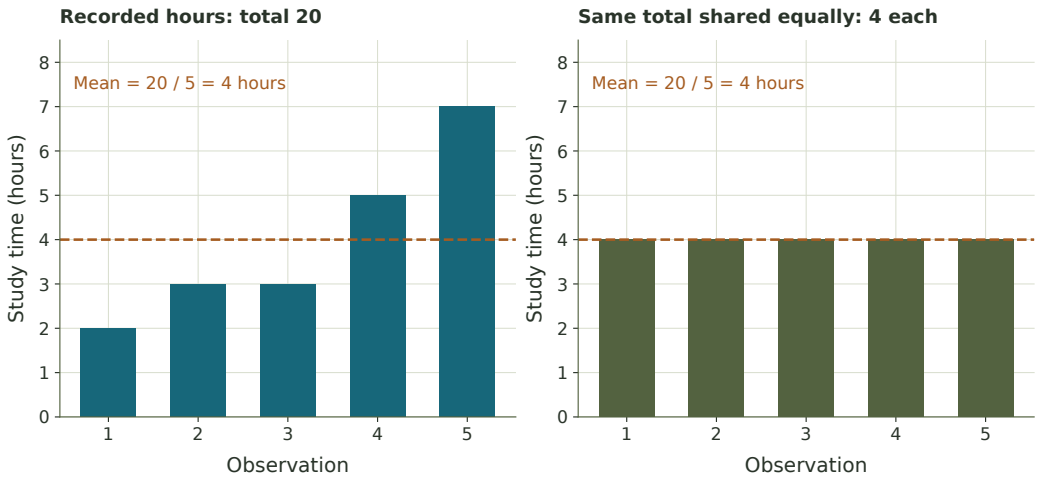
What is missing from "The average was 25"?

3.2 The mean: share the total equally

Add the values and divide by how many there are. This gives the *mean*, often called the average. Imagine sharing the total equally among all the observations.

$$\bar{x} = \frac{\text{sum of the values}}{\text{number of values}} = \frac{x_1 + x_2 + \cdots + x_n}{n}.$$

We write the sample mean as $\bar{x}$, pronounced 'x-bar'. The symbol n is the number of observations, and $x_1, x_2, \ldots, x_n$ label their values. The mean equals their sum divided by n.



The mean preserves the total: number of observations × mean = original total.

A mean does not have to appear in the list

For 2, 3, 3, 5, and 7 hours, the total is 20 and the mean is $20 \div 5 = 4$ hours. The mean of 4 hours describes the group as a whole, even though none of the observations equals 4.

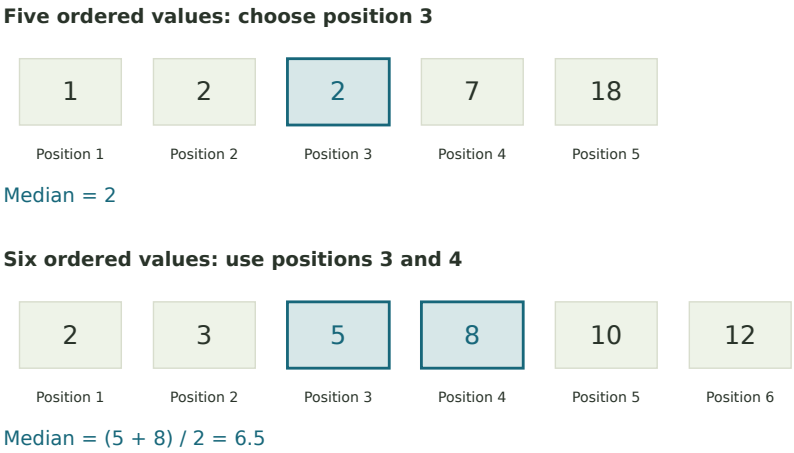
You can reverse the calculation. If four values have mean 12, their total must be $4 \times 12 = 48$. If three are 9, 11, and 13, the missing value is $48 - 33 = 15$.

Pause and try 2

The values are 4, 6, and 8 minutes. Find the mean, then recover their total from the mean.

3.3 The median: find the middle after ordering

Put the observations in order before finding the *median*. With an odd number of values, choose the middle position. With an even number, average the two middle values.



Count positions in the ordered list. The middle of the list is not necessarily halfway between the smallest and largest measurements.

Sort first

The values 9, 2, 4, 4, 7, and 5 become 2, 4, 4, 5, 7, 9. There are six observations. Positions 3 and 4 contain 4 and 5, so the median is $(4 + 5)/2 = 4.5$.

Repeated values matter. In 1, 2, 2, 2, 9, the median is 2. There are not equal numbers *strictly* below and above it because several observations equal it. The ordered-position procedure still works.

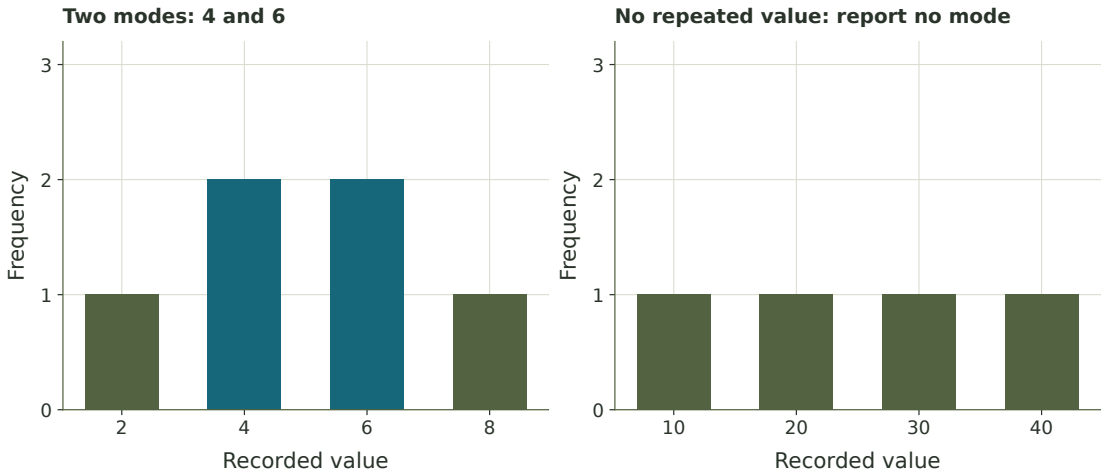
The median depends on position in the sorted list. Changing a value far above the middle may change the mean substantially while leaving the median unchanged. This can make the median useful for data with unusually large or small values.

Pause and try 3

Find the median of 3, 8, 1, and 6.

3.4 The mode: count repeated values

The *mode* is a most frequently occurring value. A frequency table makes the count visible. There can be several tied modes.



For this course, report "no mode" when every observed value appears once. A tie among repeated values can give more than one mode.

Mean, median, and mode answer different questions. The mean shares the total; the median finds the middle; the mode finds a most common value. They can agree, but they do not have to.

Use the question to choose the summary

A shop planning its most commonly requested size might care about the mode. To divide a total cost equally, the mean answers the question. To describe a middle amount when a few amounts are unusually large, the median may be more helpful.

Pause and try 4

What are the modes of 1, 1, 2, 3, and 3?

3.5 The range: measure the full width

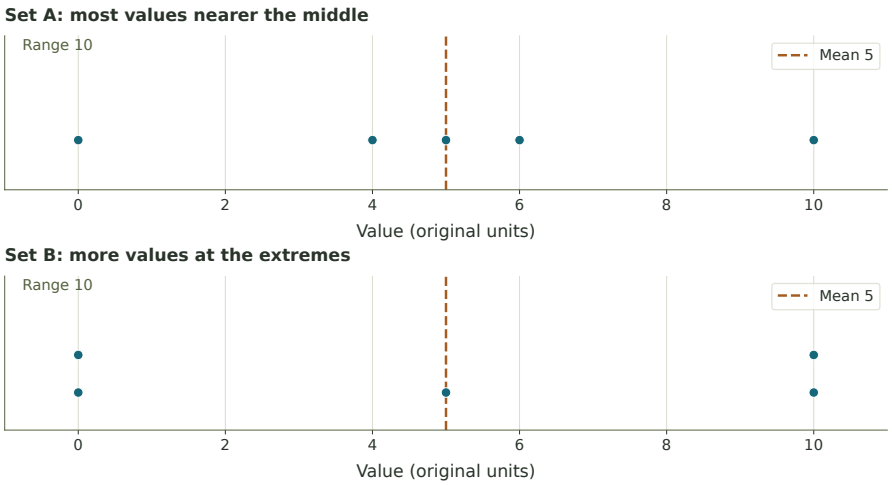
The *range* is the maximum minus the minimum. It has the same units as the observations.

$$\text{range} = \text{largest value} - \text{smallest value}.$$

Watch the subtraction sign

For -3, 1, and 4 degrees, the range is $4 - (-3) = 7$ degrees. For 8, 9, 10, 11, 12, the range is $12 - 8 = 4$. For 2, 6, 10, 14, 18, it is $18 - 2 = 16$. Those last two lists both have mean 10.

The range measures the distance between the smallest and largest observations. The graph shows how the other values are distributed. Two lists can therefore have the same mean and the same range while distributing their observations differently.



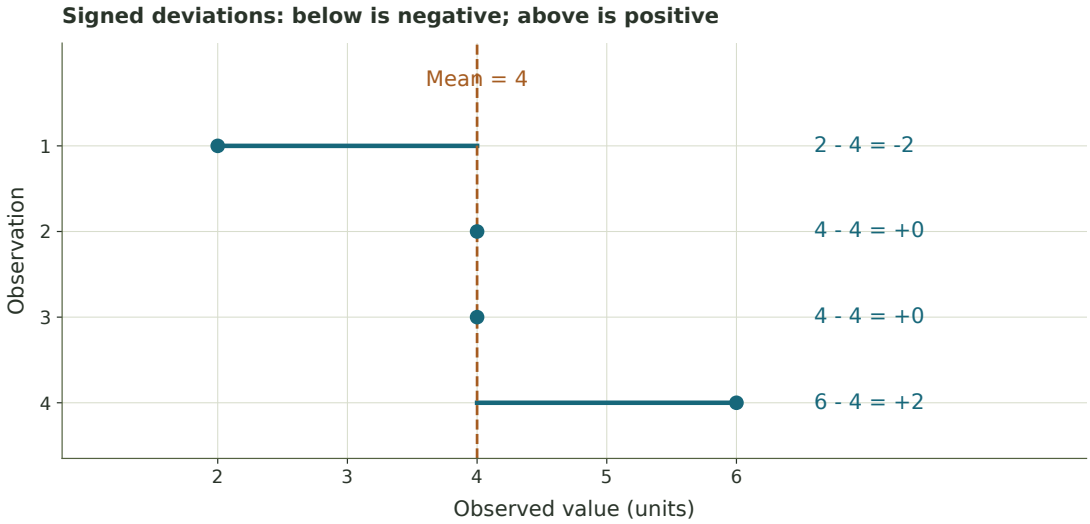
Both ranges are 10. Looking at the observations reveals information the two endpoints miss.

Pause and try 5

A data set has minimum 7 and range 18. What is its maximum?

3.6 Distances from the mean

A *deviation* is an observation minus the mean. A value below the mean has a negative deviation; one above it has a positive deviation.



The signs locate values relative to the mean. Opposite signed deviations can cancel even when the measurements vary.

For 2, 4, 4, 6, the deviations add to $-2 + 0 + 0 + 2 = 0$. The zero does not mean there is no variation. It means the mean is a balance point: the signed departures below and above it cancel.

Simply averaging those signed deviations would always give zero, apart from rounding. To describe spread, we need a calculation in which departures cannot cancel each other.

Pause and try 6

For 1, 3, and 5, find the mean and the three deviations.

3.7 Sample standard deviation, one step at a time

The *sample standard deviation*, written s, describes spread around the mean using every observation. It is useful to think of it as a scale for distances from the mean. It is not calculated by averaging the ordinary distances.

Square each deviation so opposite signs cannot cancel. Add those squares. For a sample, divide by n − 1 to obtain the *sample variance*. Finally, take the square root to return to the original units.

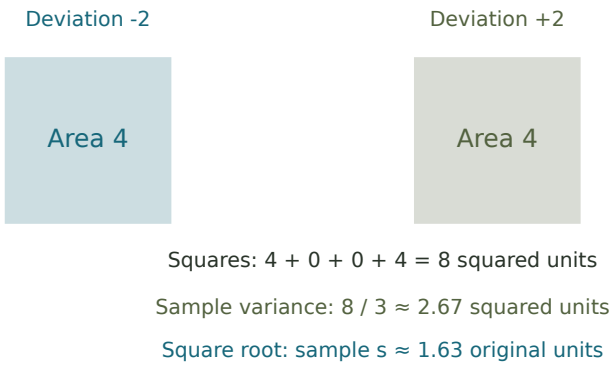
$$s = \sqrt{\frac{(x_1 - \bar{x})^2 + \cdots + (x_n - \bar{x})^2}{n - 1}}.$$

The compact symbol $\sum$ sometimes replaces the long addition. It means "add these terms." Work down the columns of a table before trying to do the whole formula at once.

Value x	Deviation x − 4	Squared deviation
2	−2	4
4	0	0
4	0	0
6	2	4
Total	0	8

Complete the last two steps

There are four observations, so n − 1 = 3. The sample variance is $8/3 \approx 2.67$ squared units. The sample standard deviation is $\sqrt{8/3} \approx 1.63$ original units. Keep the fraction until the last step; rounding the variance too early can change the answer.



Squaring changes the units. The final square root makes standard deviation comparable with the original measurements.

Why n − 1? We have already used the sample to find its mean. Once that mean and all but one deviation are known, the last deviation is fixed by their sum being zero. The sample-variance rule allows for that lost freedom. Its statistical justification concerns estimating variation in a larger population; you do not need that proof to follow the calculation here.

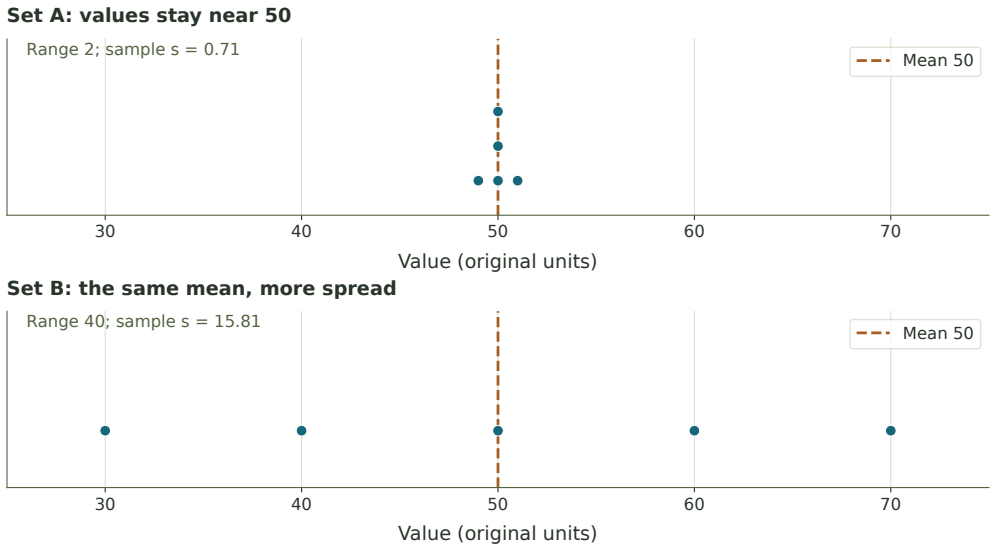
Some calculators offer both a sample SD and a population SD. A population calculation divides by the full count when the list is the entire population being described. These assigned calculations use the *sample* formula and n − 1. At least two observations are needed. With two or more identical values, s = 0.

Pause and try 7

For 1, 3, and 5, the squared deviations total 8. Find the sample variance and sample standard deviation.

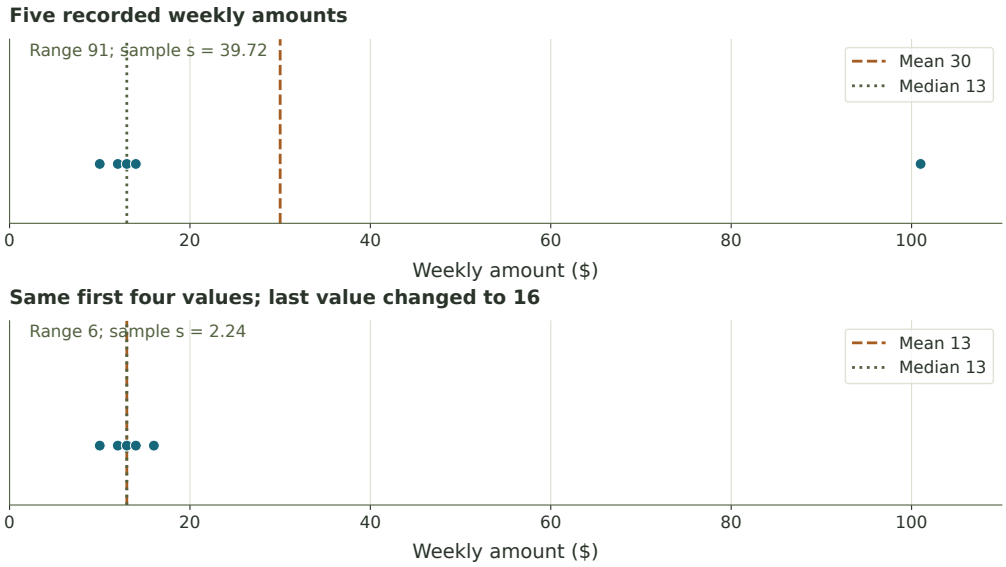
3.8 Look at the graph as well as the summaries

Compare centers first, then spreads, while keeping the setting and units in view. A larger standard deviation means greater spread around that mean. It does not automatically mean better performance.



The same axis makes the difference visible. Range and sample standard deviation both show that the second sample is more spread out.

Now consider five weekly amounts: 10, 12, 13, 14, and 101 dollars. Their mean is 30 dollars; their median is 13 dollars. The mean correctly shares the full total of 150 dollars. The median better describes the middle of these particular weekly amounts.

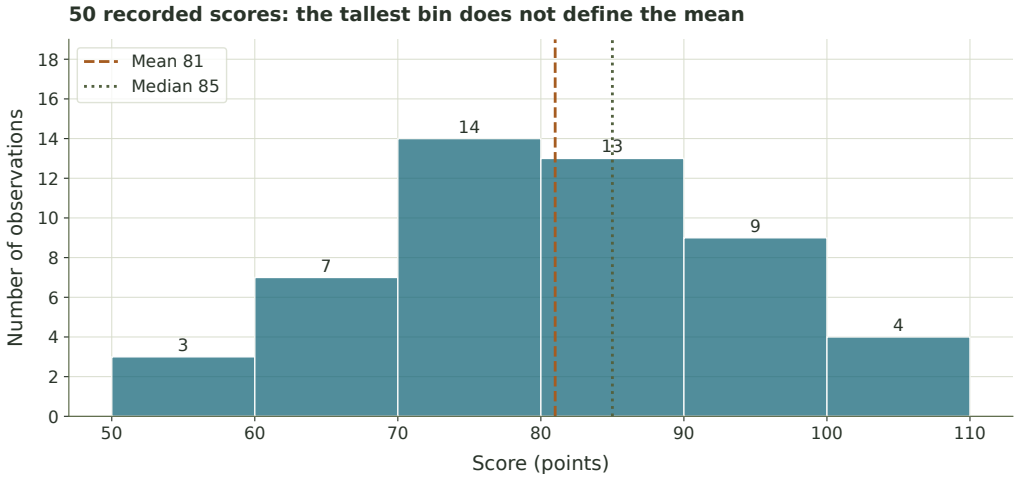


An unusually distant value can strongly change the mean, range, and standard deviation. The median is often less sensitive.

A value far from the others is worth investigating. It could be a recording error or a real unusual event. Comparing summaries with and without it shows its influence; it does not by itself justify deleting the observation.

The range is *not* resistant to extreme values. Using a median alongside a range does not make the range resistant. For this course, calculate the requested measures and explain what each reveals. Other courses may introduce measures such as the interquartile range that focus on the middle part of the data.

Histograms group observations into intervals. A bar's height counts observations in that interval. The tallest bar identifies a most common interval, not necessarily the location of the mean or median.



Here the raw scores are 55, 65, 75, 85, 95, and 105, occurring 3, 7, 14, 13, 9, and 4 times. The tallest bin contains 28% of the sample, not a majority.

Pause and try 8

If a sample has one very large value, is the range protected from its effect by reporting the median?

3.9 Weighted averages: unequal shares

In an ordinary mean, each observation receives the same share of the total weight. A *weighted average* allows different shares. Multiply each value by its weight, then add. For the form used here, the weights are nonnegative and sum to 1.

$$\text{weighted average} = w_1x_1 + \cdots + w_nx_n, \qquad w_1 + \cdots + w_n = 1.$$

Two scores with different weights

Scores of 70 and 90 have ordinary mean 80. If the first counts for 25% and the second for 75%, the weighted result is $0.25(70) + 0.75(90) = 17.5 + 67.5 = 85$. Convert the percentages to decimal weights before multiplying.

With nonnegative weights summing to 1, the result stays between the smallest and largest values. Larger weight means a value has more influence on the combined result. It does not turn the weight into a statement that the value is correct.

The attention application uses this same operation to combine numerical information. Work the familiar weighted average before reading about its role in a model.

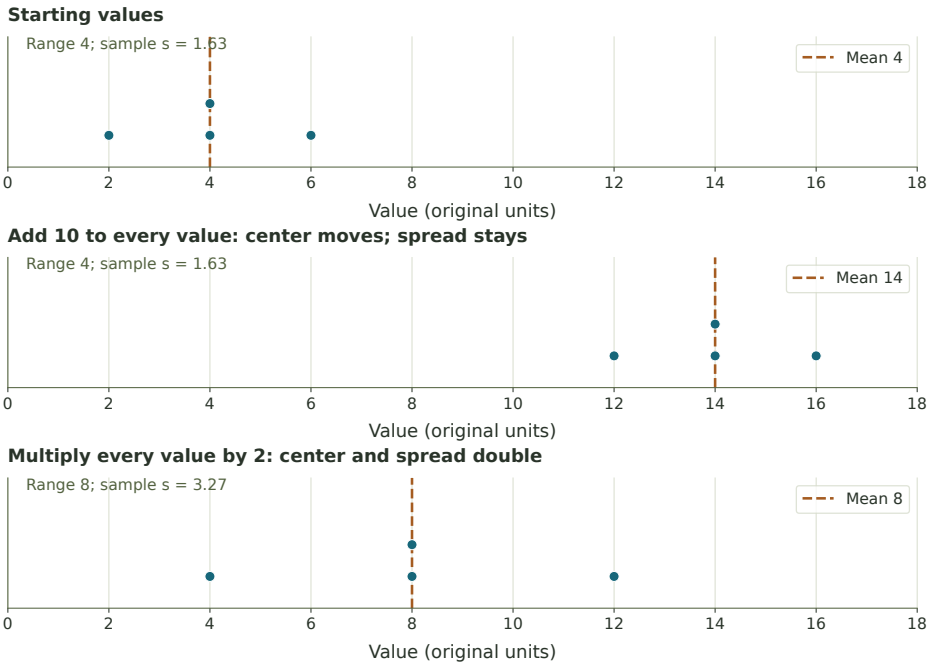
Pause and try 9

Scores of 60 and 80 receive weights 0.25 and 0.75. Find the weighted average and check its bounds.

3.10 Moving or rescaling every value

If you add the same number to every observation, all the points move together. Their mean and median move by that number, but the distances between points do not change. Range and standard deviation stay the same.

If you multiply every observation by 2, the distances double too. The mean, median, range, and standard deviation all double. Multiplication by a negative number also reverses order; spread changes by the magnitude of the multiplier and remains nonnegative.



Moving the whole data set changes its location. Stretching it changes its spread. Keep the units and transformation clear.

Pause and try 10

A sample has mean 20 and sample SD 3. Add 5 to every value. What are the new mean and sample SD?

3.11 Interpreting statistical summaries

Write the result with its units and a sentence about the data. "The five test bottles averaged 250 mL, with sample SD about 0.71 mL" tells us more than two unlabeled calculator outputs.

Question	Useful calculation	What to keep in view
What would each share be if the total were equal?	Mean	Every value affects the total
What is the middle ordered value?	Median	Sort first; use both middle values when needed
Which value repeats most often?	Mode	Ties and no repeated values are possible
How far apart are the extremes?	Range	It uses only the endpoints
How much do values vary around the mean?	Sample standard deviation	Use all values and the $n - 1$ rule here

A summary describes the observations we have. Whether those observations represent a larger group depends on how they were selected and measured. A narrow spread does not establish that the sample came from the right population or that the measuring instrument was accurate.

Week 4 introduces normal models and intervals for population quantities. Here we are learning to describe observed variation. A standard deviation alone does not prove that a data set has a bell shape or tell us a tail percentage.

Pause and try 11

Two filling machines average the target amount. Does that settle which is more consistent?

References: NIST's measures of location and measures of scale. Numerical examples in this unit are hypothetical.

Practice A: Center

A1. Find the mean, median, and mode of each data set.

(a) 4, 7, 7, 9, 11, 13, 7, 6

(b) 22, 18, 30, 22, 25, 18, 29, 22, 31

(c) 0.5, 1.2, 0.8, 1.5, 0.9, 1.1

A2. The following data records the number of customer complaints received by a store over ten days:

3, 0, 5, 2, 1, 4, 2, 8, 2, 3.

(a) Find the mean, median, and mode.

(b) On day 8, a particularly difficult situation led to 8 complaints. Describe the effect of removing this value from the data set on each of the three measures.

A3. A real estate agent reports that the mean home price in a neighborhood is \$420,000, but the median is \$285,000. In two complete sentences, explain what this difference tells a potential buyer about the distribution of home prices in the neighborhood.

A4. Seven students took a test. Six of them scored $72, 78, 84, 80, 76, 88$. The mean score for all seven students is 80. What did the seventh student score? Show your work.

Practice B: Range

B1. Find the range of each data set.

(a) 14, 22, 9, 31, 18, 27

(b) -5, 0, 3, -2, 8, -7, 1

(c) 0.4, 1.1, 0.7, 1.8, 0.3

B2. Two delivery drivers recorded the following number of deliveries per day over five days.

	Day 1	Day 2	Day 3	Day 4	Day 5
Driver A	12	14	11	13	15
Driver B	8	18	10	16	13

(a) Compute the mean and range for each driver.

(b) Which driver is more consistent? Justify your answer using the range.

B3. A data set has a range of 40 and a minimum value of 15. What is the maximum value?

Practice C: Sample standard deviation

C1. Compute the sample standard deviation of each data set. Show all steps using the deviation table.

(a) 3, 5, 5, 7

(b) 10, 20, 30, 40

(c) 6, 6, 9, 9

C2. Without computing, decide which data set has a larger sample standard deviation. Explain your reasoning.

(a) Set A: $50, 51, 50, 49, 50$ Set B: $30, 40, 50, 60, 70$

(b) Set C: $2, 2, 2, 2, 2$ Set D: $1, 2, 3, 4, 5$

C3. A data set has five values with mean $\bar{x} = 10$. The squared deviations are $4, 1, 0, 1, 4$. Compute the sample standard deviation without knowing the original values.

C4. Two machines fill bottles with juice. Machine A fills to $250, 251, 249, 250, 250$ mL. Machine B fills to $245, 255, 248, 253, 249$ mL.

(a) Compute the mean and sample standard deviation for each machine.

(b) Which machine is more consistent? Justify using the sample standard deviation.

Practice D: Interpreting summaries

D1. For each data set, compute all five statistics: mean, median, mode, range, and sample standard deviation. Use the values and statistics to explain how an extreme value affects the summaries. Which center better describes a typical observation? Explain.

(a) 12, 14, 14, 15, 16, 17, 14

(b) 5, 6, 7, 8, 9, 10, 85

D2. A meteorologist records daily rainfall (in inches) for two weeks:

0, 0, 0.2, 0, 1.4, 0, 0, 0.1, 0, 0, 2.1, 0, 0, 0.

(a) Find the mean, median, and mode.

(b) Which measure of center best represents a typical day? Justify in two sentences.

D3. A dataset has mean 20 and sample standard deviation 3. Another has mean 20 and sample standard deviation 10.

(a) Sketch a rough histogram for each, placing the center at 20.

(b) What does a larger sample standard deviation indicate about the distribution of values?

Practice E: Reading graphs

E1. The histogram below shows the distribution of heights (in inches) for a sample of 40 adults.



(a) Which interval has the largest count? Is that a majority of the observations?

(b) Is the distribution symmetric or skewed?

(c) Would you expect the mean to be close to the median? Explain in one sentence.

E2. Sketch two histograms with the same center but different spreads. Label the center on both. Below each sketch, write one sentence describing what each distribution tells an observer about the data.

E3. A data set has mean 30 and sample standard deviation 2. A second data set has mean 30 and sample standard deviation 8. An observation of 36 is taken.

(a) How many sample standard deviations above the mean is 36 in each data set?

(b) In which data set is the value 36 more unusual? Explain in one sentence.

Practice F: Weighted averages

F1. Scores 70 and 90 receive weights 0.25 and 0.75. Find the weighted result, compare it with the ordinary mean, and check that it lies between the two values.

F2. Values 2, 5, 8, 10 receive weights 0.10, 0.20, 0.60, 0.10. Check the weight total, calculate every contribution, and find the weighted result. What does the largest weight tell you?

F3. In a whole-sentence illustration, positions 2, 3, 5, 6, 8 receive weights 0.10, 0.05, 0.70, 0.05, 0.10. Find the weighted position. Explain why a decimal result does not designate a fractional word. Why would a next-token generator need to exclude words that are not available yet?

Practice G: Changing every value

G1. A sample has mean 20 and sample SD 3. Separately, (a) add 5 to every value and (b) multiply every original value by 2. Give the resulting mean and SD for each change, and explain through distances from the mean.

Chapter 3 Review

R1. The following data records the number of hours per week a sample of 10 students spend on homework:

$$4,\ 7,\ 5,\ 8,\ 6,\ 5,\ 9,\ 4,\ 6,\ 6.$$

(a) Find the mean, median, mode, and range.

(b) Construct the deviation table and compute the sample standard deviation.

(c) Describe the center and spread of this data set in two complete sentences.

R2. A company tracks the number of sick days taken by employees in one year. The data for eight employees is:

$$0,\ 1,\ 2,\ 1,\ 0,\ 3,\ 1,\ 18.$$

(a) Compute the mean and median.

(b) Which measure better represents a typical employee? Explain.

(c) If the value 18 is removed, how do the mean and median each change? Compute both.

R3. Two students each score a mean of 80 on five exams. Student A's scores are $78, 79, 80, 81, 82$. Student B's scores are $60, 70, 80, 90, 100$. Without computing, predict which student has the larger sample standard deviation. Then compute both sample standard deviations to verify.

R4. A data set has six values. Five of them are $12, 15, 14, 13, 16$. The mean of all six values is 14. Find the sixth value.

R5. A histogram of test scores is skewed left.

(a) What does it mean for a histogram to be skewed left?

(b) Would you expect the mean to be greater than, equal to, or less than the median? Explain in two sentences.

R6. The temperatures recorded over five days in two cities are:

	Day 1	Day 2	Day 3	Day 4	Day 5
City A	68	70	69	71	67
City B	55	62	74	79	85

(a) Compute the mean for each city.

(b) Compute the range and sample standard deviation for each city.

(c) Which city has more consistent temperatures? Justify using both spread measures.

R7. A quality control engineer samples 6 ball bearings and measures their diameters (in mm):

$$9.98,\ 10.01,\ 10.00,\ 9.99,\ 10.02,\ 10.00.$$

(a) Compute the mean and sample standard deviation.

(b) The specification requires that all bearings fall within 0.03 mm of the target diameter of 10.00 mm. Do all sampled bearings meet this requirement?

Solutions and Reasoning

Short check 1. We need to know what was measured, its units, which observations were included, and how much they varied. An average is a summary of a specified set of measurements.

Short check 2. The mean is $(4 + 6 + 8)/3 = 6$ minutes. Three observations $\times$ 6 minutes $= 18$ minutes, the original total.

Short check 3. Sort to get 1, 3, 6, 8. The median is $(3 + 6)/2 = 4.5$. Averaging the endpoints would give the same answer here by coincidence; the procedure uses the two middle positions.

Short check 4. Both 1 and 3 occur twice, more often than 2. The modes are 1 and 3; averaging those two modes would answer a different question.

Short check 5. Maximum $- 7 = 18$, so maximum $= 25$. Range describes a difference, not the largest value by itself.

Short check 6. The mean is 3. The deviations are -2, 0, and $+2$, which sum to zero. The observations are still different.

Short check 7. There are three observations, so divide by $3 - 1 = 2$. The variance is 4 squared units, and $s = \sqrt{4} = 2$ original units.

Short check 8. No. The median and range are separate calculations. The median can stay nearly fixed while the maximum and therefore the range change greatly.

Short check 9. The result is $0.25(60) + 0.75(80) = 15 + 60 = 75$. It lies between 60 and 80 and is closer to the more heavily weighted value 80.

Short check 10. The mean becomes 25; the sample SD stays 3. Each value and the mean moved by the same 5, leaving every deviation unchanged.

Short check 11. No. Compare their observed ranges and sample SDs, inspect the values, and apply the stated tolerance rule. Equal means can conceal very different variation.

Practice A: Center

A1. (a) Mean 8; median 7; mode(s): 7.

(b) Mean $\frac{217}{9} \approx 24.1111$; median 22; mode(s): 22.

(c) Mean 1; median 1; mode(s): no repeated value, so no mode under the course convention.

A2. The ten-day total is 30: mean 3, median 2.5, mode 2. Removing 8 leaves total 22 across nine days: mean $22/9 \approx 2.4444$, median 2, mode 2. Both the total and count change. The mode stays fixed here; a median need not remain fixed when an observation is removed.

A3. The mean is much higher than the median, suggesting some high prices are pulling the total-per-home upward. The median describes a middle observed home price; the mean describes the total divided by the count. These two summaries do not reveal the complete shape or prove how many homes are affordable to a particular buyer.

A4. $7(80) = 560$ points in all. The six known scores total 478. The missing score is $560 - 478 = 82$.

Practice B: Range

B1. (a) $31 - 9 = 22$. (b) $8 - (-7) = 15$. (c) $1.8 - 0.3 = 1.5$. The result has the original measurement units.

B2. Driver A totals 65: mean 13 and range $15 - 11 = 4$. Driver B also totals 65: mean 13 and range $18 - 8 = 10$. A is more consistent in these five recorded days under the range comparison. Equal means do not describe equal daily variation.

B3. Maximum minus 15 equals 40, so the maximum is 55.

Practice C: Sample standard deviation

C1. (a)

Value	Deviation	Squared deviation
3	-2	4
5	0	0
5	0	0
7	2	4

Mean 5. Squared-deviation sum 8. Divide by 3 for sample variance $\frac{8}{3} \approx 2.6667$. Then $s = \sqrt{\frac{8}{3}} \approx 1.6330$ in the original units.

(b)

Value	Deviation	Squared deviation
10	-15	225
20	-5	25
30	5	25
40	15	225

Mean 25. Squared-deviation sum 500. Divide by 3 for sample variance $\frac{500}{3} \approx 166.6667$. Then $s = \sqrt{\frac{500}{3}} \approx 12.9099$ in the original units.

(c)

Value	Deviation	Squared deviation
6	$\frac{-3}{2}$	$\frac{9}{4}$
6	$\frac{-3}{2}$	$\frac{9}{4}$
9	$\frac{3}{2}$	$\frac{9}{4}$
9	$\frac{3}{2}$	$\frac{9}{4}$

Mean $\frac{15}{2} \approx 7.5$. Squared-deviation sum 9. Divide by 3 for sample variance 3. Then $s = \sqrt{3} \approx 1.7321$ in the original units.

C2. (a) B has the greater sample SD: its values are farther from the common mean 50. The calculated SDs are $\sqrt{0.5} \approx 0.7071$ and $\sqrt{250} \approx 15.8114$. (b) D has greater spread. All C values equal 2, giving $s_C = 0$; D varies, with $s_D = \sqrt{2.5} \approx 1.5811$. The zero-SD statement requires at least two observations.

C3. The squared deviations total 10. With five observations, sample variance is $10/4 = 2.5$ and $s = \sqrt{2.5} \approx 1.5811$. The given squared deviations suffice because they already supply the departures from the mean.

C4. Both means are 250 mL. A has squared-deviation sum 2, giving variance $2/4 = 0.5$ and $s_A \approx 0.7071$ mL. B has squared-deviation sum 64, giving variance 16 and $s_B = 4$ mL. A is more consistent in the observed samples. This does not guarantee every future fill.

Practice D: Interpreting summaries

D1. (a) Mean $\frac{102}{7} \approx 14.5714$; median 14; mode(s): 14. Range 5; sample variance $\frac{55}{21} \approx 2.619$; sample SD $s = 1.6183$.

(b) Mean $\frac{130}{7} \approx 18.5714$; median 8; mode(s): no repeated value, so no mode under the course convention. Range 80; sample variance $\frac{18080}{21} \approx 860.9524$; sample SD $s = 29.3420$.

The second list has six values from 5 through 10 and one at 85. Its median 8 better describes that main cluster, while its mean still correctly accounts for the full total.

The range and SD are both sensitive to the distant value. Reporting a median does not make the range resistant.

D2. Rainfall totals 3.8 inches over 14 days, giving mean $3.8/14 \approx 0.2714$ inch per day. Ten values are zero, so median and mode are zero. For a typical observed day, zero conveys that most days were dry. The mean is still useful for the total-per-day rainfall over the full period.

D3. Draw both histograms centered at 20, with the first concentrated more tightly and the second extending more widely. A larger sample SD indicates greater spread around the common mean. The two summary numbers do not determine unique histograms or establish a normal shape.

Practice E: Reading graphs

E1. (a) The 66-to-below-70-inch interval has the largest count, 12. That is $12/40 = 30\%$, not a majority. (b) The bars are roughly mound-shaped and approximately symmetric, with small differences between the two sides. (c) That shape suggests the mean and median will be reasonably close. Exact raw-data means and medians cannot generally be recovered from these bin counts alone.

E2. One valid sketch uses the two data sets shown in the chapter figure: both center at 50, with one much narrower than the other. The narrow sample has values close to 50; the wide sample places observations farther from 50. A graph should label a shared measurement scale so the widths can be compared.

E3. The distances are $(36 - 30)/2 = 3$ sample SDs and $(36 - 30)/8 = 0.75$ sample SD. The observation is farther from the mean in SD units in the first data set. Without further information about shape, these distances do not supply exact tail percentages.

Practice F: Weighted averages

F1. $0.25(70) + 0.75(90) = 17.5 + 67.5 = 85$. The ordinary mean is 80. The weighted result is between 70 and 90 and closer to the more heavily weighted score 90.

F2. The weights total 1. Contributions are 0.2, 1, 4.8, and 1; their sum is 7. The ordinary mean is 6.25. Weight 0.60 makes the value 8 strongly influential in this particular combination; it does not certify that its information is correct.

F3. $0.10(2) + 0.05(3) + 0.70(5) + 0.05(6) + 0.10(8) = 0.2 + 0.15 + 3.5 + 0.3 + 0.8 = 4.95$. This is a numerical center of weighted positions, not a token at a fractional position. The complete-sentence table includes position 8 after the word at position 7. A next-token generator must mask unavailable future text; otherwise it would use information that it does not have when generating.

Practice G: Changing every value

G1. (a) The mean becomes 25 and SD stays 3: every observation and its mean move by the same 5, preserving deviations. (b) The original mean doubles to 40 and SD doubles to 6: deviations double, squared deviations quadruple, and the final square root doubles the spread measure.

Chapter 3 review

R1. Mean 6 hours, median 6 hours, mode 6 hours, and range 5 hours.

Value	Deviation	Squared deviation
4	-2	4
7	1	1
5	-1	1
8	2	4
6	0	0
5	-1	1
9	3	9
4	-2	4
6	0	0
6	0	0

Mean 6. Squared-deviation sum 24. Divide by 9 for sample variance $\frac{8}{3} \approx 2.6667$. Then $s = \sqrt{\frac{8}{3}} \approx 1.6330$ in the original units.

The sample centers at 6 weekly homework-hours, with observed values from 4 to 9 hours and sample SD about 1.63 hours.

R2. The total is 26 days, giving mean $26/8 = 3.25$ and median 1. The median better describes the middle employee in these observations. Removing 18 leaves total 8 across seven observations: mean $8/7 \approx 1.1429$ and median 1. The unusually large value substantially changes the mean, while the median stays fixed in this particular comparison.

R3. B should have greater sample SD. A has squared-deviation sum 10 and $s_A = \sqrt{10/4} \approx 1.5811$. B has sum 1,000 and $s_B = \sqrt{1000/4} \approx 15.8114$. Both means are 80, but B varies much more.

R4. Six values with mean 14 total 84. The five known values total 70. The missing value is 14.

R5. A left-skewed histogram has a longer tail toward the smaller values. Such low values tend to pull the mean toward that tail, so one generally expects the mean to be below the median in this setting. This is an interpretation of the displayed shape, not a formula for exact numerical values.

R6. City A has mean 69 degrees F, range 4 degrees F, and $s_A = \sqrt{2.5} \approx 1.5811$ degrees F. City B has mean 71 degrees F, range 30 degrees F, and $s_B = \sqrt{151.5} \approx 12.3085$ degrees F. A has the more consistent observed temperatures according to both spread measures.

R7. The mean diameter is 10.00 mm. Squared deviations total 0.001 square mm. Sample variance is $0.001/5 = 0.0002$ square mm and $s = \sqrt{0.0002} \approx 0.014142$ mm. The permitted interval is [9.97,10.03] mm. Every sampled bearing meets it; the sample alone does not establish that every future bearing will meet it.

References and activities

The NIST Engineering Statistics Handbook discusses measures of location and measures of scale. The weighted-average application uses the value-combination structure described in Vaswani and colleagues' Attention Is All You Need.

Use the explorer to compare a common axis with fitted axes and to inspect every deviation. The filling-machine application separates a target, observed consistency, and an acceptance rule. The discussion asks what an average misses in an example from your own experience. Week 4 uses these distinctions when describing normal models and uncertainty in population estimates.